

Team Details

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StateZero

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Problem Statement

Selected Problem

PS06 - Reinforcement Learning for Small Language Model (SLM) Reasoning

Problem Understanding

Current SLMs (1.5B-7B parameters) struggle with complex reasoning tasks despite recent advances in RL-based training (GRPO, DAPO, Dr. GRPO). Pure RL cannot extend a small model's reasoning frontier - it only sharpens existing sampling distributions. Key failure modes include entropy collapse, diversity loss (Pass@k degradation), length explosion, and reward hacking. The core challenge is to go beyond baseline GRPO reproduction and deliver a novel, demo-ready system that improves SLM reasoning on GSM8K, MMLU, and StrategyQA benchmarks while being feasible on a single 24-48 GB GPU. Our approach targets the unaddressed gap of adaptive compute allocation: no existing work RL-trains a separate routing model that dynamically decides whether a problem needs full chain-of-thought reasoning or a quick direct answer, paired with a tiny on-device verifier. This directly maps to Samsung Galaxy on-device deployment - the winning template from EnnovateX 2025.

Proposed Solution

On-Device System-1.5 with Distilled Tiny Verifier and RL-Trained Router

Core Idea: We build an adaptive reasoning system that pairs a DeepSeek-R1-Distill-Qwen-1.5B reasoning SLM with a tiny 400M cross-encoder verifier. The verifier acts as a difficulty router - scoring each incoming question to decide whether it needs full chain-of-thought (System-2) reasoning or a fast single-step (System-1) answer.

Three-Stage Pipeline: (1) Distill a DeepSeek-V3 question-difficulty judge into a 400M Weaver-style cross-encoder (~6 hours on 24 GB). (2) RL-train a 2-token routing head on the 1.5B reasoner using GRPO with reward = correctness - λ * tokens_used. (3) Quantize the full system to GGUF Q4_K_M for on-device deployment on Samsung Galaxy.

Samsung Mapping: The system runs entirely on-device (no cloud, no API). It maps directly to a Galaxy on-device math/coding tutor - exactly replicating the EnnovateX 2025 winning template (Team COD3INE: on-device + multi-agent + concrete Samsung product mapping).

Proposed Solution - Technical Architecture

Technical Architecture and Data Flow

Stage 1 - Verifier Distillation (~6 hrs): Distill DeepSeek-V3 question-difficulty judgments into a 400M cross-encoder using Weaver-style weak supervision. Input: question text. Output: difficulty score [0,1]. Training uses knowledge distillation from teacher ensemble (GPT-4o / DeepSeek-V3 as judge).

Stage 2 - RL Router Training (~36 hrs): Take DeepSeek-R1-Distill-Qwen-1.5B as reasoner. Add a 2-token routing head that outputs <think> or <no-think>. Train via GRPO (Unsloth + TRL GRPOTrainer + QLoRA + colocated vLLM) with reward = correctness - λ * tokens_used. ~1500 steps on a single RTX 4090 (24 GB).

Stage 3 - On-Device Deployment: Quantize both models to GGUF Q4_K_M. Inference flow: Question enters verifier (400M) which scores difficulty. If easy (score < threshold), route to single-step answer. If hard, route to full CoT with budget-forced extension (s1-style). All inference runs locally on Samsung Galaxy - zero cloud dependency.

Constraints: Single 24 GB GPU (RTX 4090). QLoRA-4bit GRPO requires ~5-7 GB VRAM for the 1.5B model at short context. Total training wall time: ~42 hours across both stages. Pinned stack: Unsloth latest, TRL 0.12+, vLLM 0.12.0-0.18.0, FlashAttention 2.

TTRL Ablation Add-on (Idea A, +1 day): Layer Test-Time RL (TTRL) with confidence-weighted voting and Clip-Cov entropy mitigation on top of the same R1-Distill-1.5B reasoner. Uses majority vote of N samples as pseudo-labels on unlabeled test data. Adds a second novelty angle within the same codebase.

Novelty & Innovation of Our Solution

State of the Art vs. Our Contribution

SOTA Baseline: DeepSeek-R1 / GRPO-based SLM training (DAPO, Dr. GRPO, Open-Reasoner-Zero) applies RL uniformly to all queries. System-1.5 Reasoning (arXiv:2505.18962) achieves 92% token reduction via adaptive shortcuts but does not RL-train a routing model. Weaver (Stanford) distills verifiers to 400M but does not integrate with adaptive compute routing.

Our Novelty (to the best of our knowledge): First system to RL-train a separate routing model on a verified correctness/latency tradeoff, combined with a tiny on-device verifier. This is a new on-device specialization pipeline - not a new RL algorithm.

Unique Value Proposition: Adaptive compute reduces average token cost by up to 92% on easy questions while preserving full CoT accuracy on hard ones. The system learns WHEN to think, not just HOW to think. Direct Samsung Galaxy deployment enables privacy-preserving, zero-latency, offline math/coding tutoring.

Pain Points Addressed: RL-trained SLMs waste compute on easy questions. On-device inference demands latency efficiency. Existing adaptive thinking methods lack RL-trained routing. No published system combines verifier distillation + adaptive routing + on-device quantization end-to-end.

Open Datasets planned to be used / published

Datasets to be Used (all open-source)

Training Datasets: GSM8K (grade school math, 8.5K train / 1.3K test), MATH-500 (competition math subset), StrategyQA (multi-hop reasoning, 2.3K questions), MMLU (multi-task language understanding, subset for unlabeled TTRL). For verifier distillation: question-difficulty labels generated by DeepSeek-V3/GPT-4o as teacher ensemble.

Evaluation Benchmarks: GSM8K test split (Pass@1, Pass@k at k=1,8,64,256), MMLU (general capability retention), StrategyQA (multi-hop reasoning), AIME24 (advanced math, 30 problems). Ablation metrics include per-token entropy heatmaps and length-distribution histograms.

Datasets to be Published: Question-difficulty annotations dataset (teacher labels from DeepSeek-V3/GPT-4o on GSM8K + MATH + StrategyQA prompts). Router training trajectories with per-question routing decisions and token usage statistics. All released under Apache-2.0 with HuggingFace dataset cards.

Open Models planned to be used / developed / trained / fine-tuned

Base Models Used (Pre-trained, Open-Weight)

Reasoner: DeepSeek-R1-Distill-Qwen-1.5B - The de-facto SLM reasoning baseline. Reaches AIME24 28.9 Pass@1, MATH-500 83.9. Fits easily on 24 GB (5-7 GB VRAM with QLoRA). Qwen2.5-Math family is exceptionally responsive to RL per SimpleRL-Zoo and 1-shot RLVR studies.

Verifier Base: A 400M cross-encoder (Weaver-style architecture from Stanford Hazy Research, June 2025). Distilled from an ensemble of weak verifiers, retaining 98.7% accuracy with 99.97% less compute than the teacher ensemble.

Models to be Developed / Fine-Tuned / Published

RL-Trained Router Adapter: QLoRA adapter for the 1.5B reasoner with a 2-token routing head. Trained with GRPO, reward = correctness - λ * tokens. Published as LoRA adapter on HuggingFace Hub with model card and Apache-2.0 license.

Distilled Difficulty Verifier (400M): Trained via knowledge distillation. Published as standalone checkpoint.

Quantized On-Device Package: GGUF Q4_K_M quantized versions of both models for Samsung Galaxy deployment.

AI/GenAI/Agentic tools used/developed

Training Infrastructure and Tools

Unsloth: Memory-efficient RL training framework. Reduces GRPO VRAM from ~510 GB to ~54 GB via linear cross-entropy loss, gradient checkpointing with async CPU offload, and shared CUDA memory pool with vLLM. Essential for single-GPU feasibility.

TRL GRPOTrainer (v0.12+): HuggingFace TRL library for GRPO training. Critic-free PPO using group-relative advantage. Combined with QLoRA (4-bit quantized LoRA adapters) for memory efficiency.

vLLM (0.12.0-0.18.0): Colocated with training for fast rollout generation during GRPO. Shares CUDA memory pool with Unsloth.

FlashAttention-2: Memory-efficient attention for long-context training and inference.

llama.cpp / GGUF: Quantization to Q4_K_M for on-device deployment. Enables inference on mobile hardware (Samsung Galaxy) with minimal RAM footprint.

WandB: Async logging for training curves, reward tracking, and ablation experiments. Integrated with tmux + auto-resume for SSH-ephemeral access reliability.

Best Practice: Frequent checkpointing (save_steps=50, save_total_limit=3), HF Hub push via hub_strategy=every_save for redundancy, deterministic seeds across 3+ runs for reproducible ablations with confidence intervals.

Additional Supporting Materials - References, Timeline, and Deliverables

Key References

DeepSeek-R1 (arXiv:2501.12948), GRPO (arXiv:2402.03300), System-1.5 Reasoning (arXiv:2505.18962), Weaver - Stanford Hazy Research (June 2025), TTRL (arXiv:2504.16084), Clip-Cov / Entropy Mechanism (arXiv:2505.22617), DAPO (arXiv:2503.14476), Dr. GRPO (arXiv:2503.20783), s1-style budget-forced reasoning (arXiv:2501.19393)

Execution Timeline (2 weeks)

Days 1-2: Environment setup, data preparation, verifier teacher label generation

Days 3-4: Stage 1 - Train 400M difficulty verifier (~6 hours training + evaluation)

Days 5-7: Stage 2 - GRPO router training on 1.5B model (~36 hours), hyperparameter sweeps

Days 8-9: TTRL ablation add-on (Idea A), benchmark evaluation across GSM8K/MMLU/StrategyQA/AIME24

Days 10-11: Quantization (GGUF Q4_K_M), on-device testing, latency/RAM/battery benchmarks

Days 12-14: Technical report writing (~8 pages), 3-min demo video recording, repo packaging (Apache-2.0, README, HF model/dataset cards, train.sh + eval.sh, pinned requirements.txt, CITATION.cff)

Expected Deliverables

GitHub repo (Apache-2.0) with full reproducibility | HuggingFace model card + dataset card | Technical report with headline benchmark number in title | 3-minute demo video (YouTube, not Drive) | On-device Pareto chart: latency vs. accuracy for fixed-CoT, no-CoT, and RL-trained adaptive router | Full ablation table: reward components, KL-beta sweep, training curves with confidence intervals over 3+ seeds